INCH-POUND
A-A-58080D
21 October 2019
SUPERSEDING
A-A-58080C
10 November 2014

COMMERCIAL ITEM DESCRIPTION

TRUCK, LIFT, FORK, ENGINE-DRIVEN, PNEUMATIC TIRES, 4,000 POUND CAPACITY AT 24 INCH LOAD CENTER

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE

1.1 Scope. This commercial item description (CID) covers the general requirements for a diesel fueled or liquid petroleum gas (LPG) fueled engine-driven, front wheel drive, rear wheel steer, pneumatic tires, 4,000 pound at 24 inch load center capacity, forklift truck. The forklift truck should be used for handling cargo in and around warehouses, loading platforms, and docks on paved or other hard surfaces.

2. CLASSIFICATION

2.1 Engine power fuel. The engine fuel systems used to power the forklifts consist of the following alternative types, as specified (see 7.3).

2.1.1 Types. The types of forklifts are as follows:

a. Type I – Diesel fuel.
b. Type II – Liquid petroleum gas fuel.

3. SALIENT CHARACTERISTICS

3.1 Safety. The operator driven forklift truck shall be capable of operation within the accuracies, limits, and specifications set forth herein. The forklift truck shall conform to ANSI/ITSDF B56.1 and Occupational Safety and Health Association (OSHA) standards in effect at the time of manufacture. An operator's seatbelt, conforming to SAE J386, in conjunction with restraint devices designed to ensure the operator's upper body remains entirely within the protection of

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the overhead guard in the event of a tip over shall be provided. The final color of the forklift truck shall be the manufacturer's standard commercial color for safe operation in an indoor environment.

3.2 Environmental requirements. The engine shall start without preheating within 5 minutes, and the truck shall operate as specified herein within 15 minutes after engine start; in any ambient temperature from 0 to 120 ºF. The forklift truck shall be capable of continuous operation as specified herein at any ambient temperature up to 120 ºF. The truck, with type C winterization, shall be capable of operating in temperatures ranging from -25 to 125 ºF.

3.3 Engine. Unless otherwise specified (see 7.3), a diesel fueled engine shall be provided.

3.3.1 Emissions. The vehicle emissions shall comply with the Environmental Protection Agency (EPA) regulations in effect at the time of vehicle manufacture. When there is a change in EPA regulations during a contract, EPA provisions shall be consulted to determine which regulations apply (e.g. TIER 1, TIER 2 or TIER 3). The vehicle shall comply with the emissions requirements of the destination state at the time of vehicle manufacture.

3.4 Fuel System. The following shall apply:

a. Type I. All essential filters, water separators, and other components required for diesel fuel operation shall be provided. It shall have a fuel tank of sufficient capacity to allow at least 8 hours of continuous operation.

b. Type II. All essential filters and components required for LPG operation shall be provided. It shall be of sufficient capacity to provide at least 6 hours of continuous operation. A means shall be provided to prevent fuel flow when the engine run control (ignition switch) is off, or if the engine should stop running.

3.5 Starting system. The engine starter motor shall not be capable of energizing while the engine is running or while the transmission is in a forward or reverse gear.

3.6 Transmission. A torque converter and a power-shift type of transmission shall be furnished. The transmission shall provide for positive inching control or declutch control that permits lifting of rated load at maximum lifting speed while the transmission control is in either forward or reverse direction, without vehicle movement. This control shall also apply to the service brakes.

3.7 Hydraulic system. A pressure relief protection device is required along with pump(s), cylinders, control valves, filter(s), reservoir, hoses, and all other components necessary to make a complete hydraulic system.

3.8 Uprights and carriage. With no load on the forks, the seated operator shall be able to see at least one fork tip at all lift heights and fork positions.

3.9 Forks. Fork dimensions shall be 1.86 inches maximum thickness, 5.06 inches maximum width, and 42 (± 0.5) inches long. Fork thickness at the tip shall be not more than 0.5 inch, and there shall be a taper on the underside of the forks of not less than 14 inches.

3.10 Fork positioner. Manually (operator) adjusted fork positioning shall be furnished. The forks shall have spacing of 12 inches (or less) when in most closed position, and 32 inches (or more) when in most spread position. These dimensions shall be measured to the outside edges of the forks.

3.11 Side shift. Hydraulically powered side shifting shall be furnished. It shall be controlled by

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seated operator, with at least 4 inches of side shift to each side of center (8 inches minimum total travel) with rated load on the forks.

3.12 Load backrest. The load backrest, in conjunction with the forks and carriage, shall provide a vertical rear load guard of at least 48 inches high, measured from the load horizontally carrying surfaces of the forks.

3.13 Steering. Power steering shall be furnished.

3.14 Service brakes. Service brakes shall be furnished. They shall conform to ANSI/ITSDF B56.1, except the maximum brake pedal force during stopping distance and drawbar drag tests shall not be more than 80 pounds.

3.15 Parking brakes. Parking brakes shall conform to ANSI/ITSDF B56.1.

3.16 Electrical system. All components necessary to make a complete electrical system shall be furnished.

3.17 Instruments. The instruments, gauges, or warning indicator lights that are on the manufacturer's standard commercial model shall be furnished. In addition, an unlighted hour meter capable of totaling at least 9999 hours shall be included.

3.18 External lights. A minimum of three floodlights shall be furnished. Two in the front, mounted on the upright, facing forward, and one mounted in the rear, facing rearward. The forklift truck shall be equipped with an amber colored strobe light with an intensity of at least 40 candela.

3.19 Taillight. One combination stoplight and taillight shall be provided.

3.20 Horn. A horn shall be furnished, with the horn actuator located in the center of the steering wheel.

3.21 Operator's overhead guard. The overhead guard shall be a falling-object-protective-structure (FOPS) conforming to ANSI/ITSDF B56.1 including the inside clearances. The height from the ground to top of the overhead guard shall not be more than 86.5 inches.

3.22 Cab. When specified (see 7.3), the forklift truck shall be equipped with a cab that shall incorporate FOPS requirements of ANSI/ITSDF B56.1. The cab shall have two opening doors and at least one fan for ventilation. There shall be sufficient window area to permit at least 250 degrees of horizontal visibility by the seated operator. The height, from the ground to the top of the cab shall be no more than 86.5 inches. A heater and defroster with fan shall be provided. There shall be at least three wipers, one on the front windshield, one on the rear window, and one on the roof window.

3.22.1 Cab extinguisher. A commercial fire extinguisher shall be mounted on the cab in an easily accessible location. The fire extinguisher shall be a minimum 2½ pound capacity ABC type, or equivalent. The fire extinguisher shall be Underwriters Laboratories (UL) listed.

3.23 Painting. The forklift truck shall be primed and painted using standard commercial practices. The final color shall be the manufacturer's standard color.

3.24 Walkway coating. Step surfaces shall be made of, or covered with, a non-slip material.

3.25 Lifting and tie down provisions. There shall be permanently installed slinging devices that enable the forklift truck to be lifted in its normal travel position, and tie downs that enable the

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forklift to be tied down to the floor of the transportation media. All devices shall be capable of accepting a chain and hook with an outside diameter of 3 inches and shall have a structural safety factor of at least 2-to-1, based on static load. The notation “LIFT HERE” and “TIE DOWN” shall be stenciled, in black, near each lifting and tie down device.

3.26 Identification plate. A corrosion resistant identification plate shall be permanently installed at an easily accessible place on the forklift truck. The identification plate shall contain the following information:

a. Nomenclature.
b. Manufacture’s Name.
c. Manufacturer’s Address.
d. Manufacturer’s Service Phone Number.
e. Manufacturer’s Model Number.
f. Manufacturer’s Serial Number.
g. Date of Manufacture.
h. Contract Number.
i. National Stock Number.
j. Registration Number.
k. Cage Code.

3.27 Performance. The forklift truck shall be capable of meeting the following performance requirements:

3.27.1 Rated load. Capability to safely handle a 4,000 pound, 48-inch-per-side cube, with the center of gravity at the center of the cube. This shall be defined, and referred to in this document as the rated load. The forklift truck shall be able to safely handle this rated load at all lift heights and fork shift positions that it is capable of attaining.

3.27.2 Load lifting speed. Shall be capable of raising rated load at a speed of at least 55 feet per minute.

3.27.3 Load lowering speed. No more than 80 feet per minute with rated load of forks and not less than 40 feet per minute with no load on the forks.

3.27.4 Right angle turn. With rated load on the forks and forklift truck located perpendicular to a wall, and with the front of the load against the wall, the forklift truck shall be able to back up and make a complete right angle turn (ending up parallel to the wall) within 170 inches.

3.27.5 Travel speed. While carrying the rated load on the forks, the travel speed shall be 10 miles per hour minimum, in both forward and reverse.

3.27.6 Slope ascension. With the forklift truck facing up the slope and with rated load on the forks, the forklift truck shall be able to accelerate up a 20% (11 degree) slope from a dead stop.

3.27.7 Upright tilt. With no load on the forklift truck, the forks shall have at least 3 degrees of forward tilt, and at least 6 degrees of rear tilt.

3.27.8 Lowest upright height (collapsed mast height). The dimension from ground to top surface of the uprights, with uprights vertical, shall not exceed 86.5 inches maximum with no load on the forks.

3.27.9 Maximum fork lift height. The maximum fork “lift height” shall be at least 144 inches

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with rated load on the forks and the forks horizontal when measured from ground to top surfaces of the forks.

3.27.10 Free lift height. The height that the forks can raise the rated load before the collapsed mast height is increased shall be at least 45 inches.

3.27.11 Drift of load. With hydraulic fluid at ambient temperatures, the lift assembly shall hold rated load at maximum lift height for at least 10 minutes with not more than 1¾-inch of vertical drift, and not more than 1 degree of rotational drift.

3.27.12 Stability. The forklift truck shall meet the ANSI/ITSDF B56.1 "forward stacking," "forward travel," "lateral stacking," and "lateral travel" stability requirements.

3.27.13 Noise limits. The maximum allowable noise level shall not exceed 92 decibel A (dB(A)) at the operator's ear. If the noise level at the operator's ear exceeds 84 dB(A) (85 to 92 dB(A)), a warning sign stating that operator’s hearing protection is required shall be installed inside the cab clearly visible to the operator. The sign shall be at least 4 inches square, with a yellow background.

3.27.14 Electromagnetic interference (EMI). The forklift truck shall comply with the EMI requirements shown in table I.

Table I. Electromagnetic interference.

Name	Description	Parameters
SAE J551/1	Compatibility	N/A
CISPR 12 & CISPR 25	Emissions	(Frequency Range: 30 to 1000 MHz)
MIL-STD-461, RS-103	Susceptibility	(Frequency Range: 100 kHz to 18 GHz) 20 V/m (100 kHz to 200 MHz) 50 V/m (above 200 MHz)

3.28 Backup alarm. An audible backup alarm shall be furnished. The alarm shall automatically activate when the transmission selector is placed in reverse.

3.29 Winterization. When specified (see 7.3), the forklift truck should be furnished with type C winterization that shall protect to -25 ºF and shall consist of the following:

a. Starting aid. Glow plug(s) or a measured-shot, ether injection system shall be furnished.

b. Cab. The cab shall meet the requirements of 3.22, and shall also contain thermal insulation, if necessary, to meet the heating requirements of 3.29(c).

c. Cab heater and defroster. The heater shall have sufficient capacity to maintain a temperature of at least 40 ºF at cab floor level in an ambient temperature of -25 ºF.

d. Antifreeze. Engine coolant shall be protected to -25 ºF with antifreeze.

3.30 Tires. The tires shall be pneumatic, and tire loading shall not exceed those specified in the Tire and Rim Association (TRA) Yearbook. When specified (see 7.3), the tires shall be solid-pneumatic.

3.31 Tropical rust proofing. When specified (see 7.3), tropical rust proofing shall be provided.

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This should be considered a premium rust proofing package usually consisting of an additional coating. The forklift truck shall be treated with a corrosion preventative compound complying with A-A-59295. Many of these vehicles are used in tropical/coastal environments and experience high humidity and salt spray. Therefore, they usually exhibit high levels of corrosion in the areas listed below. At a minimum, these areas shall be coated:

a. Cab, interior.
b. Cab, exterior.
c. Seams.
d. Welds.
e. Hidden recessed areas.
f. Locations where dissimilar metals come in contact.
g. Area above fuel tank(s).
h. Engine oil pan and transmission oil pan.
i. Radiator support structure.
j. Battery box.

4. REGULATORY REQUIREMENTS

4.1 Recycled, recovered, environmentally preferable, or bio-based materials. Recycled, recovered, environmentally preferable, or bio-based materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR). However, used, rebuilt, or remanufactured components, pieces, and parts shall not be incorporated in the forklift truck.

4.2 Green Procurement Program (GPP). The GPP is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. The GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100% of EPA designated product purchase that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired: a) competitively within a reasonable timeframe; b) meet appropriate performance standards, or c) at a reasonable price. The prime contractor is responsible for ensuring that all subcontractors comply with this requirement.

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product Conformance. The forklift truck provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices and be the same forklift truck offered for sale in the commercial market. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

5.2 Market acceptability. The following market acceptability criteria are necessary to document the quality of the product to be provided under this CID.

5.2.1 Commercial. The vehicle furnished shall be, as of the date of award, the latest model (with minor modifications) of an established product previously produced and sold in substantial commercial quantities for at least three years. The vehicle furnished, or earlier models of it,

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absent minor modifications, shall have routinely been supported with spare/repair parts which were produced or sold in the normal course of business. At the time of delivery, the contractor shall furnish the Administrative Contracting Officer with verification of compliance with these requirements. Offers shall provide, as part of their proposals, the following information on the vehicle (or its earlier models) being offered as a commercial item:

a. Total sales to the U.S. Government or to contractors for U.S. Government use during the three years.
b. Total sales of the item to the general public during the three years.
c. Length of time the item has been sold in the commercial market place.

6. PACKAGING

6.1 Packaging. Preservation, packing, and marking shall be as specified in the contract. Unless otherwise specified in the contract, the packaging shall be according to normal commercial practice and packed to assure acceptance by common carrier and provide product protection against loss and damage during multiple shipment, handling, and storage. Shipping containers shall be properly marked and in compliance with both national and uniform motor freight classifications.

7. NOTES

7.1 Intended use. The forklift truck is intended for use in handling 4,000 pound capacity cargo in and around warehouses, loading platforms, and docks on paved or other hard surfaces.

7.2 Source of documents.

7.2.1 Federal Acquisition Regulations (FAR). Copies of FARs may be obtained from the Superintendent of Documents, Government Printing Office, Washington, DC 20402, or online at https://www.acquisition.gov/?q=browsefar.

7.2.2 Government documents. Copies of specifications, standards, and handbooks are available from DLA Document Services, 700 Robbins Avenue, Building 4/D, Philadelphia, PA 19111-5094, or online at https://quicksearch.dla.mil/.

7.2.3 Industrial Truck Standards Development Foundation (ITSDF) standards. Copies of ITSDF standards may be obtained from Industrial Truck Standards Development Foundation, Suite 460, 1750 K Street NW, Washington, DC 20006, or online at http://www.itsdf.org/.

7.2.4 Occupational Safety and Health Association (OSHA). Copies of OSHA documents may be obtained from Occupational Safety and Health Association, 200 Constitution Avenue, NW, Washington, D.C. 20210, or online at http://www.osha.gov/.

7.2.5 Society of Automotive Engineers (SAE) standards. Copies of SAE standards may be obtained from the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096, or online at http://www.sae.org/.

7.2.6 Tire and Rim Association (TRA) standards. Copies of TRA standards may be obtained from the Tire and Rim Association, Inc., 175 Montrose West Avenue, Suite 150, Copley, OH 44321, or online at http://www.us-tra.org/publications.html.

7.2.7 Underwriters Laboratories (UL) standards. Copies of UL standards may be obtained from Underwriters Laboratories INC., 333 Pfingsten Road, Northbrook, IL 60062, or online at http://www.ul.com/.

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7.3. Ordering data. When specified the following shall be furnished:

a. A liquid petroleum gas (LPG) fuel system, if required (see 3.3 & 3.4).

b. Cab, when specified (see 3.22 & 3.22.1).

d. Winterization C (see 3.29).

e. Tires (see 3.30).

f. Tropical rust proofing (see 3.31).

7.4 National Stock Number (NSN). The following is a list of NSNs assigned that correspond to this CID. The list may not be indicative of all possible NSNs associated with this CID.

NSN	3930-01-433-0885

7.5 Keywords.

Cargo
Front wheel drive
Hydraulic
Paved surfaces
Rear wheel steer

7.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

MILITARY INTERESTS:

Custodians:
Army - AT
Navy - SA
Air Force - 184

Preparing Activity:
Air Force - 184

Agent:
Air Force - 110

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